

Integrated Teaching in Anatomy

- Integrated teaching program on a particular topic/organ /organ system should be organized in each term. The topics which are related should be prepared after discussion with the teachers of Anatomy/Physiology & Biochemistry. The horizontal process of Integrated teaching program will help the students to have a simultaneous views of different aspects of Anatomical/Physiological & Biochemical details of a particular topic/organ /organ system.

Topics	Learning Objectives	Term	Department
1. Cell	Students will be able to ● describe the structure & functions of different constituents of cell ● explain membrane transport, membrane potentials & action potentials	I	Anatomy and dental anatomy Physiology & Biochemistry
2. Heart	Students will be able to ● describe the gross anatomy & clinical anatomy of heart ● describe the structure & properties of cardiac muscle ● describe the phases & events of cardiac cycle	I	Anatomy and dental anatomy Physiology & Biochemistry
3. Alimentary system	Students will be able to ● describe the gross anatomy & clinical anatomy of alimentary system ● describe the movements, transport & mixing of food in GIT ● explain the mechanism of regulation of secretion of digestive juices	I	Anatomy and dental anatomy Physiology & Biochemistry

Topics	Learning Objectives	Term	Department
4. Mouth cavity & Salivary glands	Students will be able to ● describe boundaries, contents of mouth cavity ● describe location , nerve supply, mode of termination of ducts of salivary glands and their functions	II	Anatomy and dental anatomy Physiology& Biochemistry
5. Composition and functions of different parts of nervous tissue.	Students will be able to ● describe& classify neurons & neuroglia ● explain the functions of neurons & neuroglia	II	Anatomy and dental anatomy Physiology& Biochemistry
6. Autonomic Nervous System	Students will be able to ● describe& classify autonomic nervous system ● explain the functions of different parts of autonomic nervous system	II	Anatomy and dental anatomy Physiology & Biochemistry

Teaching / Learning & Assessment Methods

Teaching / Learning Method	Teaching Aid	In Course Assessment	Summative Assessment
Lecture	Computer & multimedia Slide projector, overhead projector (OHP), writing board.	Item Examination: Oral, Practical	- Written - Oral - Practical
Regional Anatomy: Demonstration & Tutorial	Cadavers, prosected parts, bones, viscera and other specimens of body parts, models, charts, writing board, Illustration sheets/posters, OHP, video, slide projector, computer with CD ROM, radiographs & other images.	Card Completion Examination Written, Oral Practical	
Regional Anatomy: Dissection	Computer & multimedia Cadavers, prosected parts, specimens and bones.	a) Term Examinations: Written, Oral Practical	
Cell Biology & Histology Tutorial & Practical	Microscope, slide projector, OHP, , Illustration sheets (including photomicrographs & drawings)/posters, video projector, computer with CD ROM drive	Preparation of exercise book	

Integrated Teaching in Physiology and Biochemistry

Integrated teaching program on a particular topic/organ /organ system will be organized in each term. The topics which are related should be prepared after discussion with the teachers of Anatomy/Physiology/Biochemistry. The horizontal process of Integrated teaching program will help the students to have a simultaneous views of different aspects of Anatomical/Physiological and Biochemical details of a particular topic/organ /organ system.

TOPICS	LEARNING OBJECTIVES	TERM	DEPARTMENT
1. Cell	Students will be able to - describe the structure & functions of different constituents of cell - explain membrane transport, membrane potentials & action potentials - state the composition of ECF & ICF compartments	I	Anatomy Physiology & Biochemistry
2. Heart	Students will be able to - describe the gross anatomy & clinical anatomy of heart - describe the types & regulation of blood pressure - describe & interpret the cardiac markers	I	Anatomy Physiology & Biochemistry
3. Lung	Students will be able to - describe the gross anatomy & clinical anatomy of Lungs - describe the spirometry & its clinical application - describe the regulation of respiration	I	Anatomy Physiology & Biochemistry
4. Hepatobiliary system	Students will be able to - describe the gross anatomy & clinical anatomy of hepatobiliary system - interpret the liver function test & explain its clinical importance - explain the role of liver in metabolism	II	Anatomy Physiology & Biochemistry
5. Kidney	Students will be able to - describe the gross anatomy & clinical anatomy of kidney - explain the mechanism of urine formation - interpret kidney function test - explain the role of kidney in regulation of water, electrolytes & acid base balance	II	Anatomy Physiology & Biochemistry
6. Thyroid & Parathyroid gland	Students will be able to describe - the gross anatomy & clinical anatomy of thyroid & parathyroid gland - the hormones of thyroid & parathyroid gland : biosynthesis , transport, functions , mechanism of action & regulation of secretion - hypo & hyperthyroidism - tetany - thyroid function tests & their interpretation	II	Anatomy Physiology & Biochemistry